

```

        (rmse, mae, r2) = eval_
metrics(test_y, predicted_qualities)

        print("Elasticnet model
(alpha=%f, l1_ratio=%f):" % (alpha,
l1_ratio))

        print(" RMSE: %s" % rmse)
        print(" MAE: %s" % mae)
        print(" R2: %s" % r2)

        mlflow.log_param("alpha", alpha)
        mlflow.log_param("l1_ratio",
l1_ratio)
        mlflow.log_metric("rmse", rmse)
        mlflow.log_metric("r2", r2)
        mlflow.log_metric("mae", mae)

        tracking_url_type_store =
urlparse(mlflow.get_tracking_uri()).
scheme

        # Model registry does not work
with file store
        if tracking_url_type_store !=
"file":

            # Register the model
            mlflow.sklearn.log_
model(lr, "model", registered_model_
name="ElasticnetWineModel")
            else:
                mlflow.sklearn.log_model(lr,
"model")

```

The code can be executed with the default hyper-parameters:

```
python sklearn_elasticnet_wine/train.py
```

The hyper-parameters can be supplied as arguments as well:

```
python sklearn_elasticnet_wine/train.py
<alpha> <l1_ratio>
```

Every time the code is executed, the information is logged in the folder `mlruns`.

Comparing the models with the UI

The trained models can be compared by

launching the UI with the following command:

```
mlflow ui
```

The MLflow UI is shown in Figure 6.

Packaging training code

With the MLflow projects conventions, the dependencies and the entry point

Date	User	Source	Version	Parameters		Metrics		
				alpha	l1_ratio	mae	r2	rmse
2018-06-04 23:00:10	mlflow	train.py	05e956	1	1	0.649	0.04	0.862
2018-06-04 23:00:10	mlflow	train.py	05e956	1	0.5	0.648	0.046	0.859
2018-06-04 23:00:10	mlflow	train.py	05e956	1	0.2	0.628	0.125	0.823
2018-06-04 23:00:09	mlflow	train.py	05e956	1	0	0.619	0.176	0.799
2018-06-04 23:00:09	mlflow	train.py	05e956	0.5	1	0.648	0.046	0.859
2018-06-04 23:00:09	mlflow	train.py	05e956	0.5	0.5	0.628	0.127	0.822
2018-06-04 23:00:09	mlflow	train.py	05e956	0.5	0.2	0.621	0.171	0.801
2018-06-04 23:00:09	mlflow	train.py	05e956	0.5	0	0.615	0.199	0.787
2018-06-04 23:00:09	mlflow	train.py	05e956	0	1	0.578	0.288	0.742
2018-06-04 23:00:09	mlflow	train.py	05e956	0	0.5	0.578	0.288	0.742
2018-06-04 23:00:09	mlflow	train.py	05e956	0	0.2	0.578	0.288	0.742
2018-06-04 23:00:08	mlflow	train.py	05e956	0	0	0.578	0.288	0.742

Figure 6: MLflow UI

Run 7c1a0d5c42844dcd8f5191146925174

Experiment Name: Default Start Time: 2018-06-04 23:47:22

Source: train.py Git Commit: 3aa48cffe58b8d9d69f5

User: mlflow Duration: 145ms

▼ Parameters

Name	Value
alpha	0
l1_ratio	0

▼ Metrics

Name	Value
mae	0.578
r2	0.288
rmse	0.742

► Tags

▼ Artifacts

- model
 - MLmodel
 - model.pkl

Full Path: /Users/mlflow/mlflow-prototype/mlruns/0/7c1a0d5c42844dcd8f5191146925174/artifacts/model/MLmodel
Size: 259B

```

artifact_path: model
flavors:
  python_function:
    data: model.pkl
    loader_module: mlflow.sklearn
    sklearn:
      pickled_model: model.pkl
      sklearn_version: 0.19.1
run_id: 7c1a0d5c42844dcd8f5191146925174
utc_time_created: '2018-06-05 06:47:22.757025'

```

Figure 7: Model artifacts from UI

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